

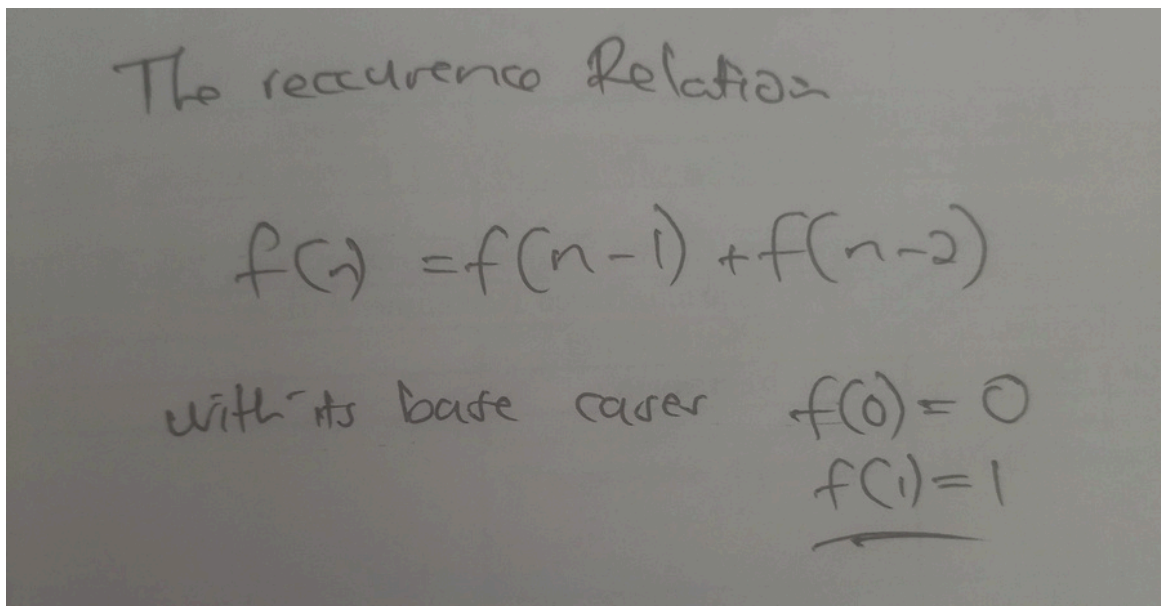
ASSIGNMENT SUBMISSION

1. What is meant by optimal substructure in dynamic programming?

Optimal substructure means that the optimal solution to a problem can be constructed from the optimal solutions of its smaller subproblems. In other words, solving subproblems optimally leads to the overall optimal solution.

Example: In the shortest path problem, the shortest path from A to C through B includes the shortest path from A to B.

2. Write the recurrence relation for computing the nth Fibonacci number using dynamic programming.



The recurrence Relation

$$f(n) = f(n-1) + f(n-2)$$

with its base cases

$$f(0) = 0$$
$$f(1) = 1$$

3. State one major difference between memoization and tabulation in dynamic programming.

Memoization is a top-down approach that solves problems recursively and stores results to avoid recomputation.

Tabulation is a bottom-up approach that solves subproblems iteratively and stores results in a table.